Web Scraping, EDA, and Data Visualization Report

Made By: Seemoan

Task 1: Web Scraping:

- **Objective:** To extract book titles and prices from a website and store them in a CSV file for analysis.
- **What I Did:**
  1. Opened the webpage containing book data.
  2. Inspected HTML elements to locate Title and Price.
  3. Wrote Python code using requests to fetch HTML content.
  4. Used BeautifulSoup to parse HTML and extract required data.
  5. Stored extracted data in books.csv

- **How I Did It:**

1. Imported libraries:

```
import requests
from bs4 import BeautifulSoup
import pandas as pd
```

2. Fetched HTML from the website:

```
response = requests.get(URL)
soup = BeautifulSoup(response.text, 'html.parser')
```

3. Extracted Title and Price using HTML tags and class selectors.

4. Created a pandas DataFrame and saved it as books.csv.

Task 2: Exploratory Data Analysis (EDA)

Objective: To explore the dataset, understand its structure, detect patterns or anomalies, and prepare for visualization.

- **What I Did:**
  1. Loaded books.csv into a pandas DataFrame.
  2. Explored dataset using head(), info(), and describe().
  3. Cleaned Price column by removing £ signs and converting to float.
  4. Checked for missing values and duplicate titles.
  5. Calculated statistics: max, min, and average prices.

- **How I Did It:**

1. Imported libraries:

```
import pandas as pd
```

2. Loaded CSV and inspected data:

```
df = pd.read_csv('books.csv')
print(df.head())
print(df.info())
```

3. Cleaned Price column:

```
df['Price'] = df['Price'].str.replace('£','').astype(float)
```

4. Analyzed data:

```
print(df.describe())
print('Most expensive book:', df['Price'].max())
print('Cheapest book:', df['Price'].min())
print('Average price:', df['Price'].mean())
print('Missing values:', df.isnull().sum())
print('Duplicate titles:',
df.duplicated(subset=['Title']).sum())
```

Task 3: Data Visualization

- **Objective:** To visualize data for better understanding, insights, and decision-making.

- **What I Did:**

1. Created visualizations for price distribution, top books, and price spread.
2. Used Matplotlib and Seaborn to generate charts.
3. Enhanced visuals with titles, labels, colors, and gridlines.

- **How I Did It:**

1. Imported libraries:

```
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Set Seaborn style:

```
sns.set_style('whitegrid')
```

3. Created charts:

- Histogram of book prices:

```
sns.histplot(df['Price'], bins=10, color='orange')
plt.title('Price Distribution of Books')
plt.xlabel('Price')
plt.ylabel('Number of Books')
plt.show()
```

- Bar chart for top 10 most expensive books.

- Boxplot to visualize outliers.